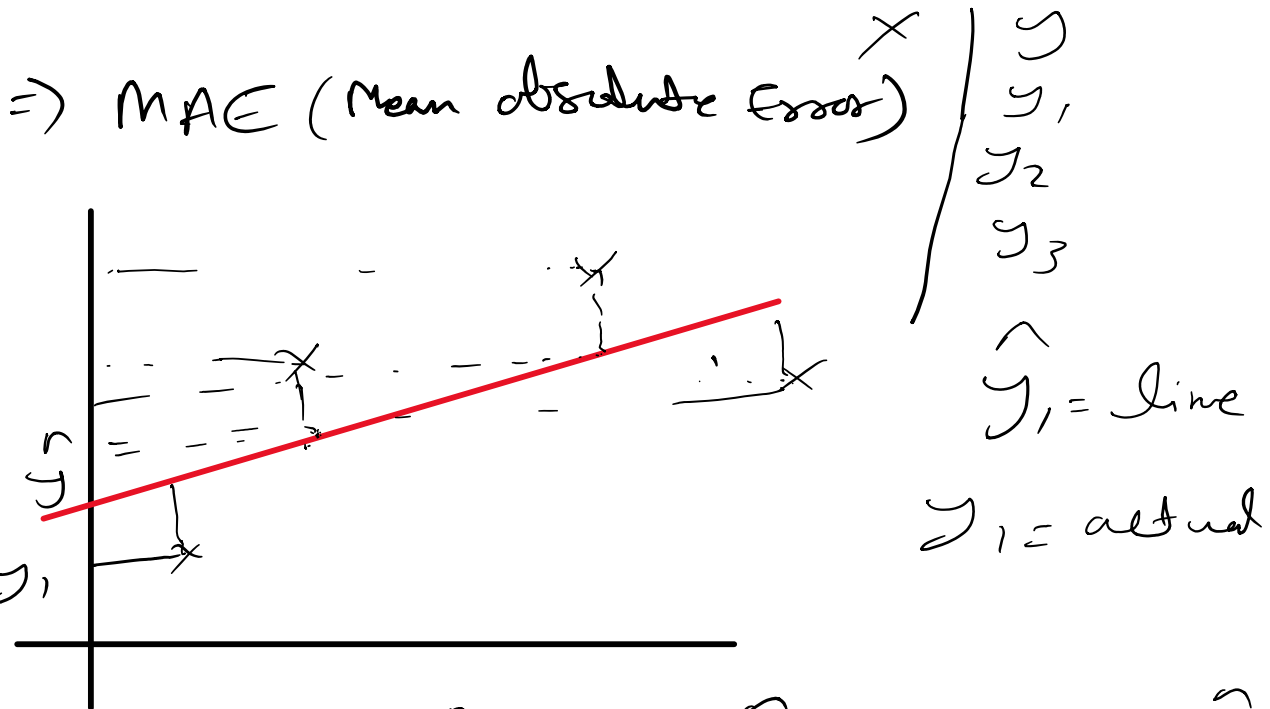




$$\Rightarrow \frac{|y_1 - \hat{y}_1| + |y_2 - \hat{y}_2| + \dots + |y_n - \hat{y}_n|}{n}$$

$$\Rightarrow \boxed{\text{MAE} = \frac{\sum_{i=1}^n |y_i - \hat{y}_i|}{n}}$$

$\Rightarrow$ Advantages :-

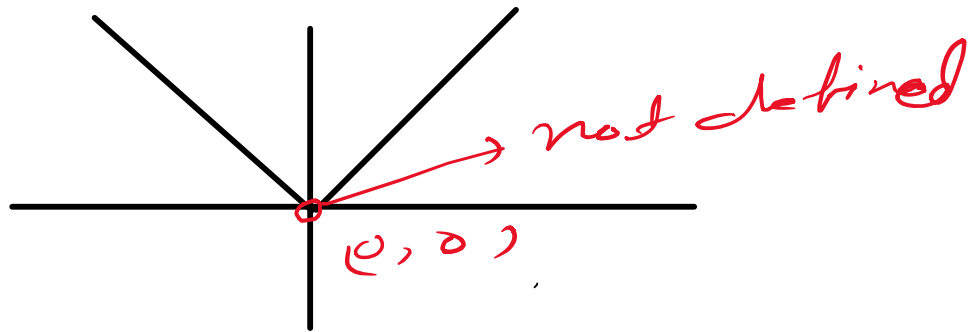
① Same Unit

② Robust to outliers

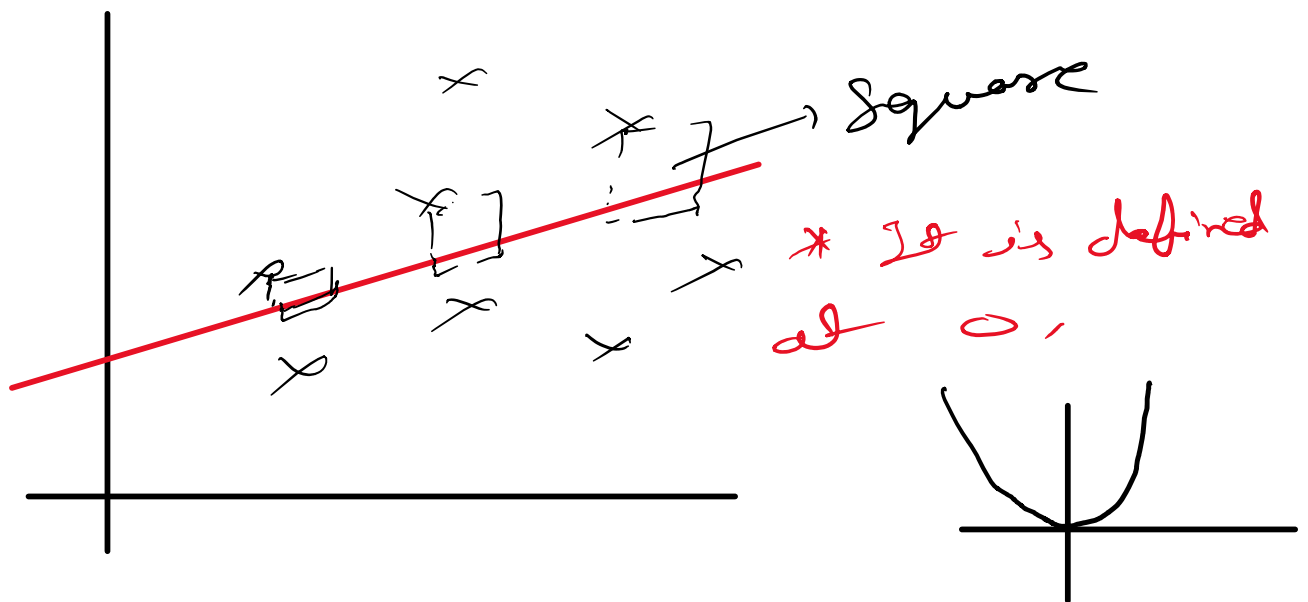
It handles the outliers.

⇒ Disadvantage:-

* It is not defined at zero



⇒ MSE (mean Squared Error)



$$MSE = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}$$

⇒ forget → try to minimize them

⇒ Advantage : —

() loss function

⇒ Disadvantage :-

() Unit is in Square :-

() If the outlier is present then

() not Robust to outliers

⇒ RMSE (Root Mean Square Error)

$$\Rightarrow \text{RMSE} = \sqrt{\text{MSE}}$$

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}}$$

⇒ Advantage :-

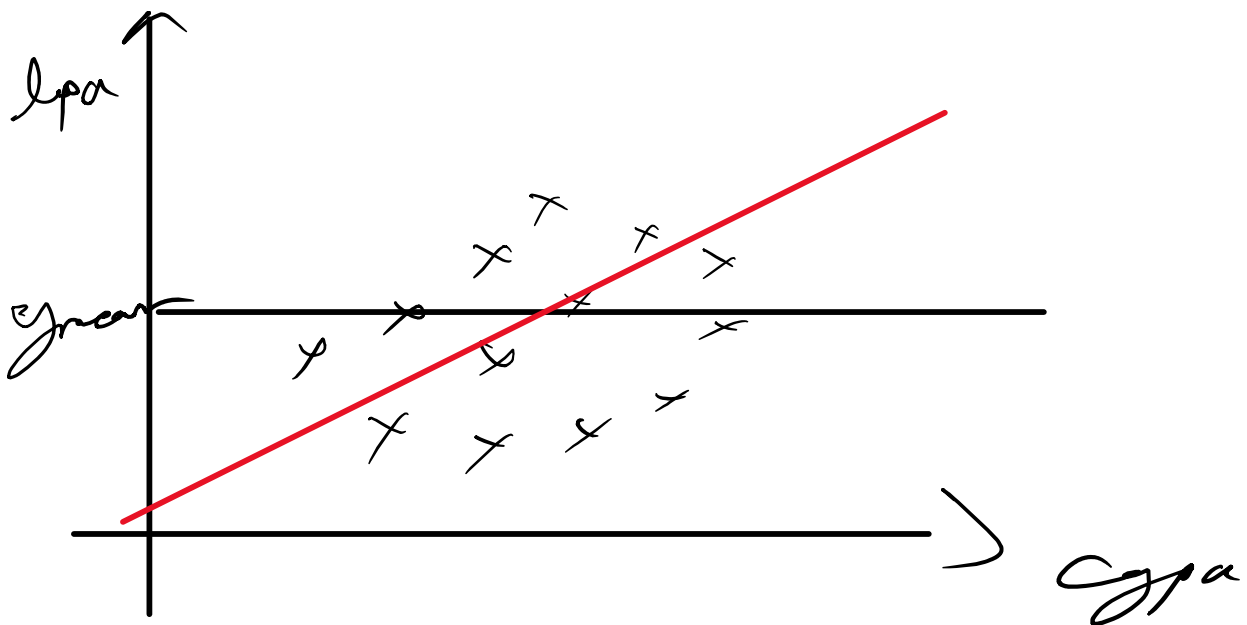
() Same unit as the column unit.

⇒ Disadvantage

() ^{not} Robust for outliers

⇒ R² Score :-

C_{gpa} | $package(lpa)$



$$R_2 = 1 - \frac{SSR}{SSM}$$

$$R_2 = 1 - \frac{\left[\sum_{i=1}^n (y_i - \hat{y}_i)^2 \right]_{\text{reg}}}{\left[\sum_{i=1}^n (y_i - \bar{y})^2 \right]_{\text{mean}}}$$

If $R_2 = 0$ → worst

$R_2 = 1$ → best

⇒ $R_2 = \text{neg}$

$$SSR > SSM$$

⇒ Adjusted R_2 score :-

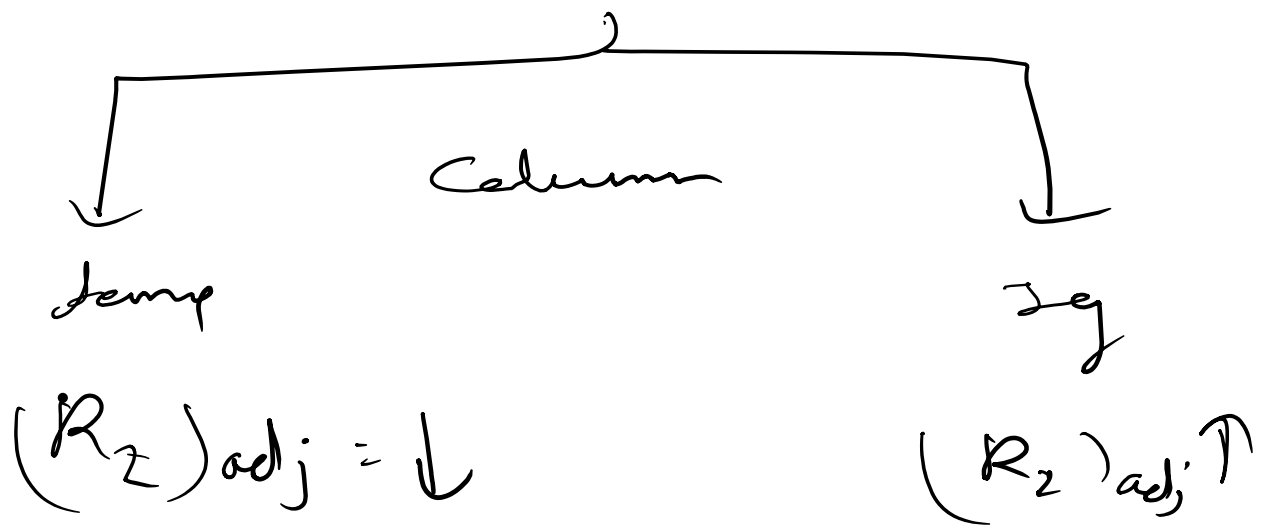
$$(R_2)_{\text{adj}} = 1 - \frac{(1 - R_2)(n - 1)}{n - k - 1}$$

$$\left[\frac{\quad}{(n-1-k)} \right]$$

$$R_2 \rightarrow$$

$n \rightarrow$ no. of rows

$k \Rightarrow$ no. of columns



$\Rightarrow$ Multiple linear Regression